Question			Answer	Marks	AO	Guidance	
13	(a)		When $t = 10$ $\mathbf{r} = \begin{pmatrix} 500 \\ 280 + 50 - 280e^{-1.6} \end{pmatrix}$ Distance $= \lvert\mathbf{r}\rvert = \sqrt{500^2 + 273.47^2}$ $= 569.9$ m	**M1** **M1** **A1** **[3]**	**3.4** **3.1b** **1.1**	Finds displacement as vector or in two directions separately Pythagoras' used Must be scalar answer	Soi Allow use of **i** and **j** vectors throughout
13	(b)		$\mathbf{v} = \begin{pmatrix} 50 \\ 5 + 44.8e^{-0.16t} \end{pmatrix}$	**M1** **A1** **A1** **[3]**	**3.4** **1.1** **1.1**	Attempt to differentiate at least one component One correct component or scalar in one direction Both correct components in a vector	soi
13	(c)		[For large values of t, $e^{-0.16t} \approx 0$] So velocity tends to $\mathbf{v} = \begin{pmatrix} 50 \\ 5 \end{pmatrix}$	 **B1** **B1** **[2]**	 **3.4** **2.2a**	 Either value seen Allow for vector or both components clear which is horizontal and which vertical	 Do not allow if the final answer is scalar
13	(d)		The model should take into account a resistance in the horizontal (x) direction.	**B1** **[1]**	**3.5c**	Must indicate the resistance has a horizontal component.	Do not accept "air resistance" on its own. SC1 "Factor e^{-kt} should be included" or similar